

Breathing Inequality: Air Pollution and Environmental Justice in NYC

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The Question

New York City is one of the most studied urban environments in the world, and its air quality has measurably improved over the past decade and a half. But averages hide things. When we started looking at pollution across individual neighborhoods, a more complicated picture emerged, and that is that not every community has benefited equally, and the neighborhoods carrying the heaviest pollution burden are often the ones with the fewest resources to deal with its consequences.

We set out to answer three questions. Can we predict pollution levels across NYC neighborhoods using historical patterns and geographic features? Which neighborhoods consistently face the highest exposure? And is that exposure correlated with race and income in ways that go beyond coincidence?

Where the Data Comes From

Our air quality data comes from the NYC Environment and Health Data Portal, which tracks pollution readings across 42 neighborhood health districts (called UHF42 areas) from 2009 through 2023. We focused on three pollutants that are directly tied to traffic and combustion: PM2.5 (fine particulate matter), NO2 (nitrogen dioxide), and ground-level Ozone.

To answer the environmental justice question, we pulled demographic data directly from the US Census Bureau's American Community Survey using their public API, race, ethnicity, median household income, and poverty rates for every census tract in the five boroughs. We then mapped those tract-level figures up to the UHF42 neighborhood boundaries so they could be matched with our pollution profiles.

What We Found: Predicting Pollution

We tested three models: a simple linear regression to establish a baseline, then Random Forest and Gradient Boosting, both of which learn non-linear patterns from the data. One of our most important methodological decisions was how we evaluated them. A common mistake with time series data is to randomly split observations into training and test sets, which lets the model effectively learn from the future. We deliberately avoided that by training only on data through 2022 and holding out all of 2023 as a true test.

That decision exposed something significant. The linear regression looked reasonable on the surface, scoring an R^2 of 0.81 on the hold-out set. But when we applied proper time-respecting cross-validation, that score collapsed. It was not actually predicting pollution. It was memorizing the future. Which won't help us nearly as much as others. The tree-based models held up, both Random Forest and Gradient

Boosting reached R^2 values of 0.98 under rigorous evaluation, with average prediction errors of roughly 1 ppb for NO₂ and 1 $\mu\text{g}/\text{m}^3$ for PM_{2.5}.

The feature importance analysis told us something useful about what actually drives these predictions. By far the most important inputs were the lag features: what pollution looked like last season and the season before that. Geography and year matter, but recent history matters most. The neighborhoods that were dirty in 2009 are largely still the dirtiest in 2023, just at lower absolute levels across the board.

The one pollutant neither model handled as well was PM_{2.5}. Fine particle concentrations spike unpredictably during wildfires, temperature inversions, and localized industrial events. Those spikes do not appear in historical patterns, so a model trained only on past readings will always miss them. That is a genuine ceiling, not a modeling failure we could tune away.

What We Found: Who Bears the Burden

We used K-Means clustering to group the 41 neighborhoods we had complete data for into four groups based on their combined demographic and pollution profiles. The results were clear.

The cluster with the highest concentration of Hispanic residents, centered on Bronx neighborhoods like Fordham, Crotona, and Hunts Point, had the lowest median household incomes in the dataset (around \$25,000 to \$30,000), poverty rates above 40%, and above-average NO₂. These communities sit at the intersection of high exposure and low capacity to respond. That is the compounded burden that environmental justice research has documented for decades: it is not just that pollution is higher, it is that the resources to buffer its health effects are also lower.

The counterpart of our findings is that some of the wealthiest neighborhoods in Manhattan, places like Gramercy Park, Chelsea, and Lower Manhattan, record the highest absolute NO₂ readings in the city. That is entirely a function of traffic density. But high income changes what those readings mean in practice. Better-sealed apartments, air conditioning, healthcare access, and the ability to simply spend less time outdoors during bad air days all insulate residents from the health consequences. The pollution number and the health burden are not the same thing for communities at different ends of the income distribution.

Why It Matters

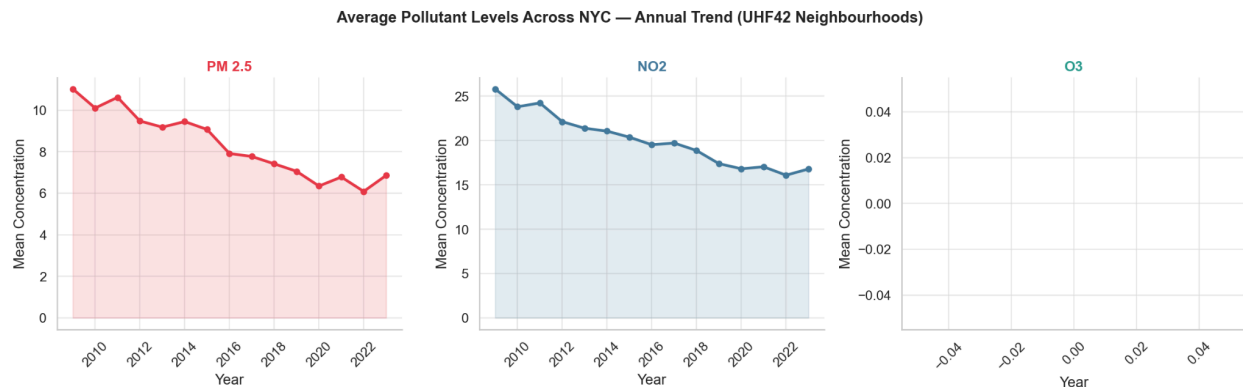
The patterns we found are not random and they are not new. They reflect decades of land use decisions: where highways were routed, where industrial facilities were sited, where transit investment went and where it did not. Data science cannot undo those decisions, but it can make the consequences visible in a way that is hard to dismiss.

There are real limits to what we built. Our model cannot predict the next wildfire spike. The Census demographics smooth over five years and miss rapid gentrification (the process of wealthier taking over poorer areas). And clustering neighborhoods by race and income carries its own risks: the goal is to hold systems accountable, not to reduce communities to a data point. We tried to be careful about that distinction.

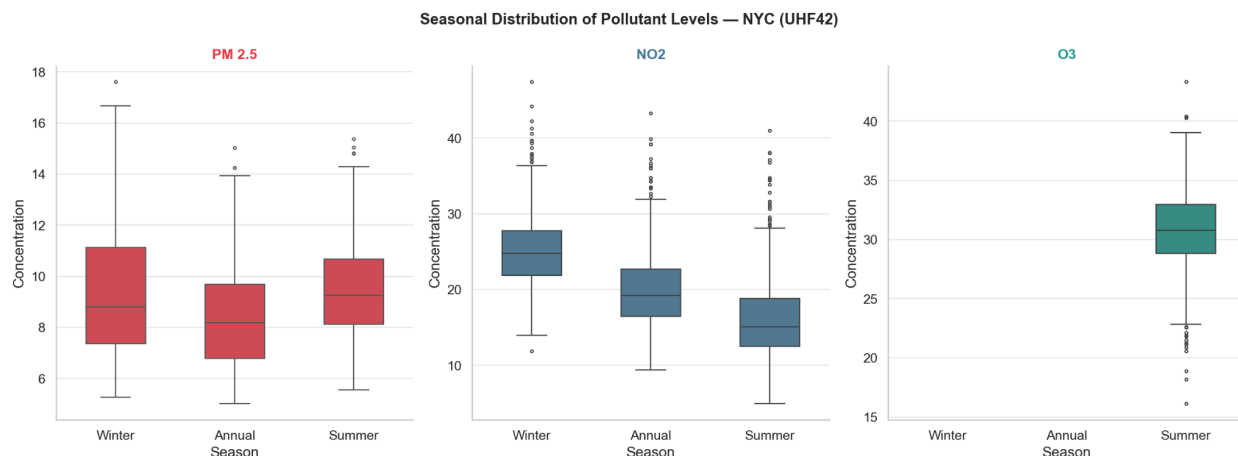
What we can say is this: air quality in New York City is improving. But improvement distributed unevenly is still inequity. The neighborhoods that started highest are still highest, and the people living in them are still the ones with the least margin for error when the air gets bad.

Visuals Explanation

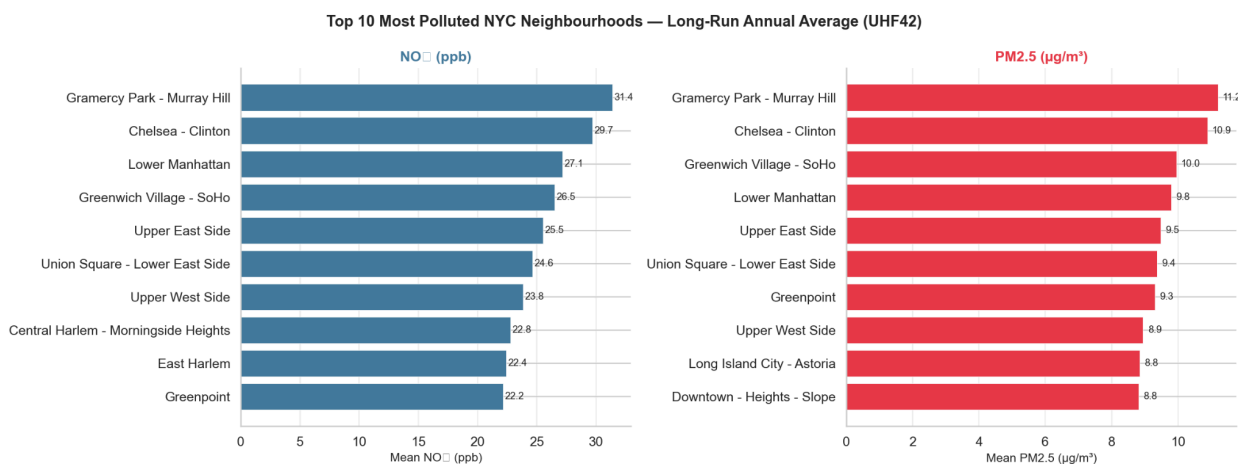
Average Pollutant Levels Across NYC - The line graphs tracks the 14-year journey of PM_{2.5} and NO₂, revealing a clear, non-random downward trend. This visual confirms that city-wide air quality has generally improved, which serves as a foundational baseline for the study. By showing this long-term improvement, it helps the reader understand that while general policy interventions are working, we must look deeper into the data to see if these health benefits are being distributed equitably across all 42 districts.



Seasonal Distribution of Pollutant Levels - The box plots illustrate that the time of year is a major factor in pollutant concentration, showing how Ozone peaks during the summer while NO₂ remains higher during winter. This insight is vital because it proves that air quality risks are not static; the "problem" changes with the weather. This helps us conclude that public health warnings and city resources must be seasonal to effectively protect residents from the specific respiratory risks associated with each time of year.

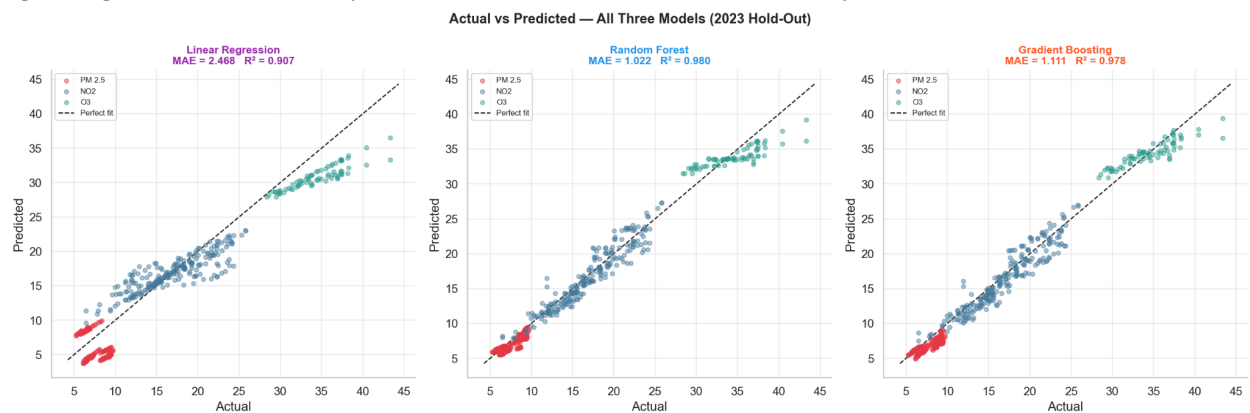


Top 10 Most Polluted NYC Neighbourhoods - The bar charts rank the areas most affected by pollution, specifically identifying locations like Gramercy Park and Chelsea. This visual directly answers the research question regarding whether some neighborhoods consistently experience higher exposure than others. By naming these hotspots, the data moves from abstract city-wide averages to actionable information that can be used to prioritize localized public health interventions.

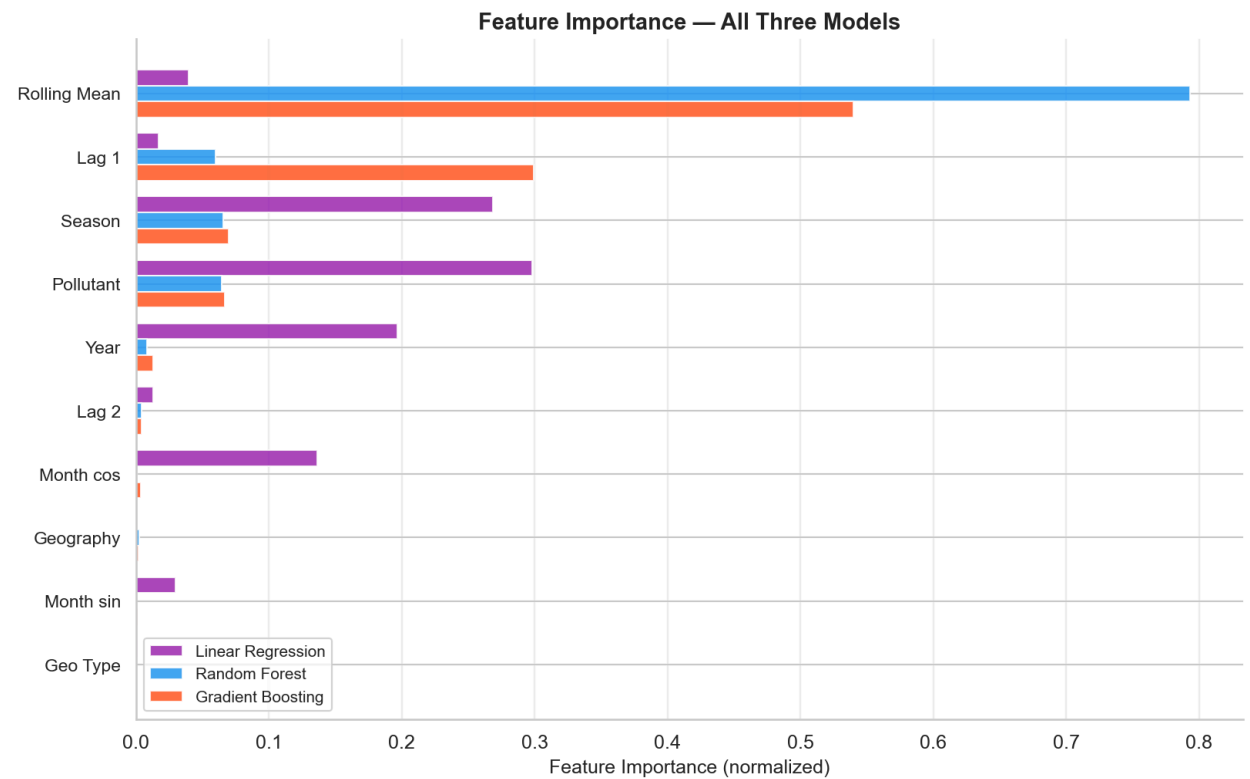


Actual vs Predicted - The scatter plots show how well each model performed, visually supporting the conclusion that these models are reliable for predicting future air quality. Because the points follow a

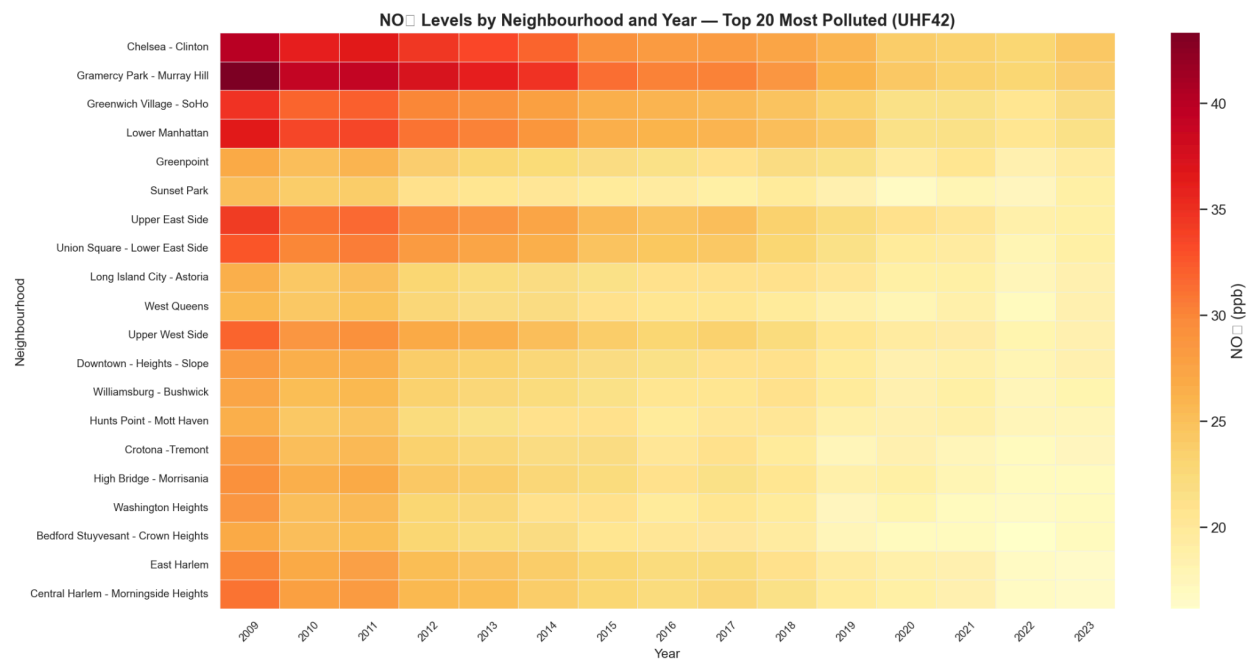
tight diagonal line, the visual provides evidence of the model's accuracy.



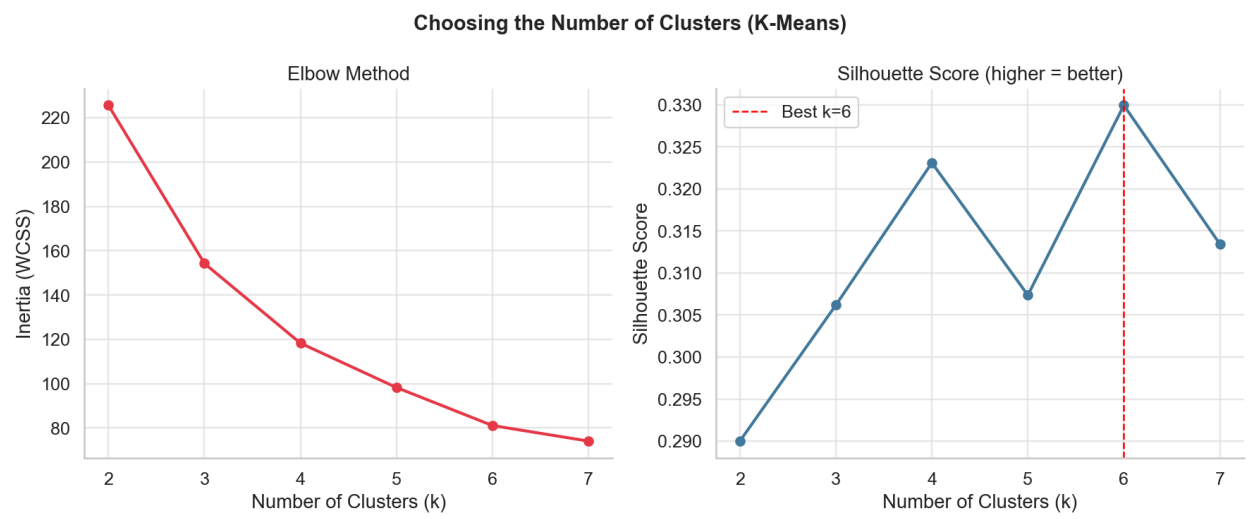
Feature Importance - The feature importance chart demonstrates that variables like "rolling mean" and "pollutant type" are the strongest predictors compared to simple geographic features. This finding is significant because it shifts the investigation from where the pollution is to why and when it occurs. This helps understand the underlying drivers of air quality, suggesting that recent history and seasonal factors are the keys to accurate forecasting.



NO₂ Levels By Neighbourhood and Year - The NO₂ heatmap provides a bird's-eye view of all 42 districts, highlighting which neighborhoods consistently face the highest NO₂ levels. As the colors shift from red to yellow over time, we can identify "sticky" areas where pollution persists longer than in the rest of the city. This helps identify the persistence of pollution, proving that a resident's long-term health outcome is significantly tied to their specific geographic location.



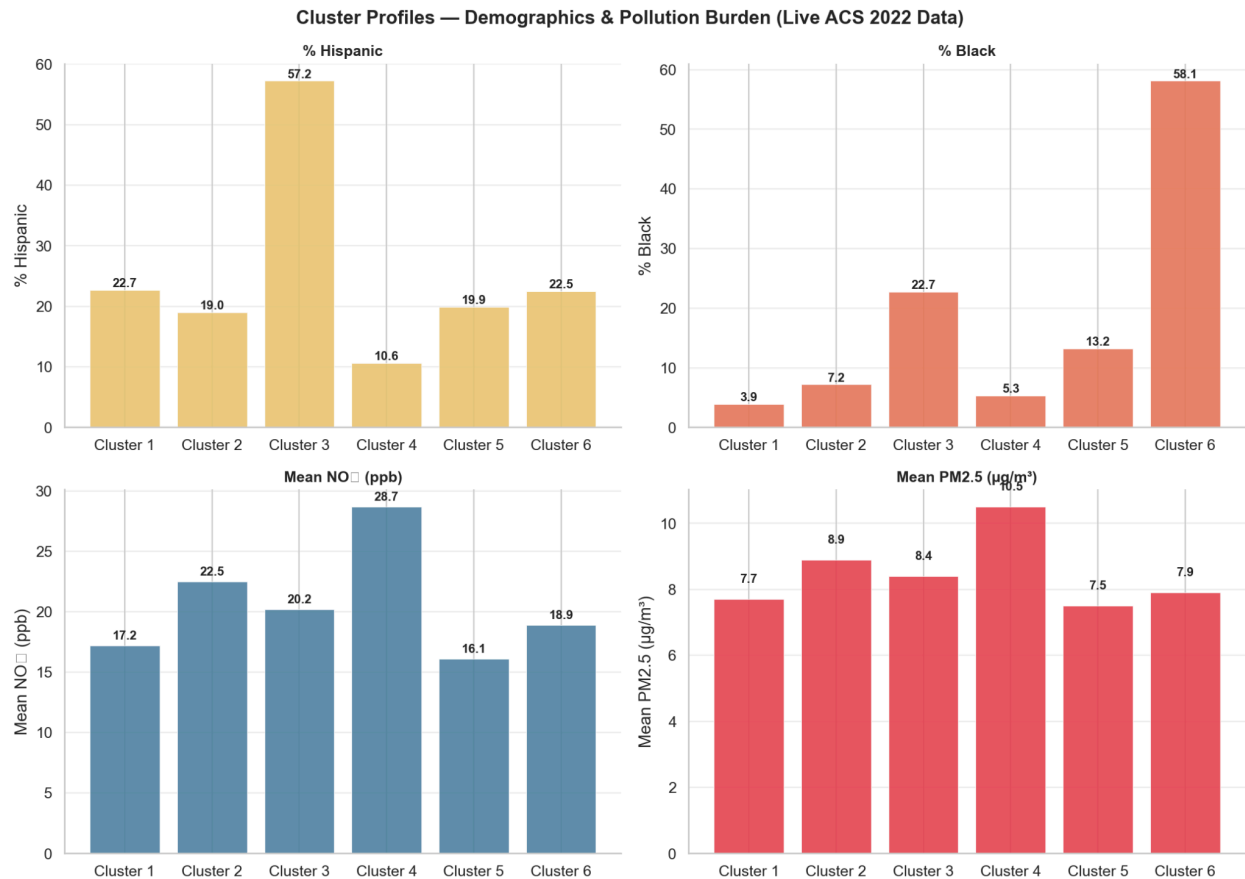
Choosing the Number of Clusters (K-Means) - In the Elbow and Silhouette graphs show the mathematical reasoning behind selecting 7 clusters to represent NYC. This proves that the groupings were not random but were chosen based on statistical rigor to best capture the diversity of NYC's urban environment.



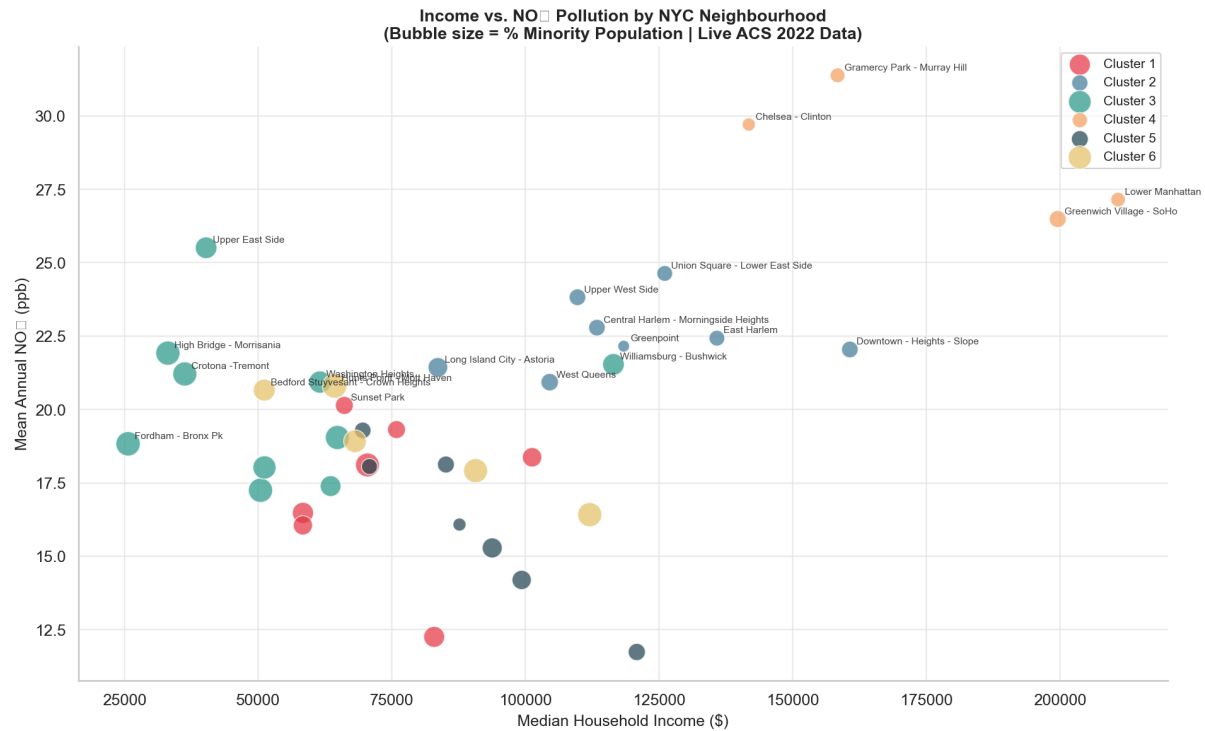
Environmental Justice - Pollution Burden vs. % Minority Population - The scatter plots compare pollution burden versus the percentage of minority residents to directly address fairness concerns. These visuals show a relationship between a neighborhood's demographic makeup and its pollution levels, highlighting potential inequalities in exposure. This is critical, as it transforms the project from a purely technical analysis into a study of social equity and environmental justice.



Cluster Profiles - Demographics & Pollution Burden - The cluster profile bar charts summarize the types of neighborhoods identified by the machine learning models. They show how different clusters vary in both demographics and pollution burdens, such as PM_{2.5} and NO₂. This allows for a more organized discussion of the data, moving beyond individual neighborhoods to identify shared "profiles" of risk that can help policymakers design targeted solutions for specific types of communities.



Income vs. NO₂ Pollution by NYC Neighbourhood - The bubble chart brings in socioeconomic factors by plotting median household income against pollution levels, with bubble size representing minority populations. This explores the "who" of the problem, showing that income and race are significant predictors of who is most affected by exposure.



References

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